

Ho Chi Minh City, October 25, 2022

COURSE SYLLABUS APPLIED CALCULUS FOR IT

COURSE CODE: 501031

1. Information:

No of credits:	4(3,1)				ECTS:	6.08
Time allocation:	Theory (hours):	45	Practice (hours):	30	Self-study (hours):	120
Prerequisite:	No				Prerequisite code:	No
Prior-Completion:	No				Prior-Completion code:	No
Co-requisite:	No				Co-requisite code:	No
Programme:	Have 2 programs				Programme code:	F7480103, F7480101
Year:					Semester:	
Type Of Course:					Language Of Instruction:	
Mode Of Delivery:					Name Of Lecturer(s):	

2. Course objectives:

No.	Course Objectives (COs)
1	Remember the basic concepts of calculus and their applications.
2	Understand the knowledge of calculus and its applications.

3	Apply calculus knowledge
4	Apply programming on calculus

3. Course learning outcomes (CLOs):

No.	CLOs	PLOs
1	Describe the concepts of calculus, their application, especially in the field of computer science.	F7480103-PLO5,F7480101-PLO5
2	Explain the calculus concepts, their meanings, calculations and applications in the scientific and technological language.	F7480103-PLO5,F7480101-PLO3
3	Apply the calculus knowledge to solve specific problems, especially problems in computer science.	F7480103-PLO5,F7480101-PLO1
4	Analyze the problems in terms of the calculus knowledge.	F7480103-PLO1,F7480101-PLO2
5	Evaluate the pros and cons of the calculus algorithms.	F7480103-PLO1,F7480101-PLO3

4. Brief course content:

Key topics of the course include: functional calculus and applications, numerical and exponential series, function extremes, differential equations, integral calculus, optimization, and some basic calculus methods.

The course includes theory lectures and computer practice using Matlab

5. Student's tasks:

Attendance:

- + Students attend all the class sessions (theory, practice, discussion). Students who are late or absent more than a specified number of days (notice will be given depending on the semester) will not allowed to take the final exam.
- + Students build the learning plan of the subject, learn with the spirit of initiative, eager to learn, serious attitude.
- + Students learn, improve the knowledge of the subject with relevant documents at the library of Ton Duc Thang University.

Full participation in class activities:

- + Students need to be active, cooperative, actively discuss in teamwork.
- + Students need to be active, self-discipline, answer questions or argue in the direction of the lesson constructing.

Complete homework assignments:

- + Students should read in advance and review the materials after each lesson, while actively reading and expanding the reference materials to complete and improve the knowledge of the course.
- + Students have to complete the assigned assignments fully and on time.

6. Teaching materials:**– Textbook:**

[1]. Maurice D. Weir, Joel Hass, George B. Thomas, [2010], Thomas' calculus, Pearson Education, Boston.

– Supplementary Readings:

[2]. R. L. Burden, J. D. Faires, [2011], Numerical Analysis, 9th edition, Brooks/Cole, Boston

[3]. James Stewart, [2012], Calculus, Brooks/Cole, Belmont.

– Additional readings:

[4]. R. W. Hamming, [1986], Numerical methods for scientists and engineers, Dover, New York.

[5]. Steven C. Chapra, [2012], Applied numerical methods with MATLAB for engineers and scientists, McGraw-Hill Education, New York.

[6]. Timothy A. Davis, [2011], MATLAB primer, CRC Press, Boca Raton.

7. Description of Evaluation:

Evaluation categories	Weight (%)	Types of questions	CLO1	CLO2	CLO3	CLO4	CLO5
Process evaluation 1	30	Process Exercise	X	X			
Process evaluation 2	20	Process Exercise	X	X	X		
Mid-term test	20	Essay			X	X	
Final examination	50	Report			X	X	X

8. Schedule:

Session	Content	Organization of teaching				Self-study	CLOs	Requirement	Student's tasks
		T	E	P	D				
	Chapter 1: Functions, limits and continuity	6	0	12	0	24			
1 - 2	- Set - Function - One to one functions - Inverse functions - Straight line and Tangent - Definition of Limit - Definition of Continuity - Limits of Rational Functions - Limits of Algebraic Functions - One-Sided Limits and Continuity - Jump Discontinuity - Continuity of a Function on an Interval - Squeeze Theorem - Limit at Infinity - Intermediate Value Theorem Teaching methods: Lecture, examples, exercises	6	0	12	0	24	[1], [2], [3], [4]		At class - Positive listening, asking questions - Practice as instructed - Doing exercises After class - Do homework Read: - [1] chapter 2 - [4] chapter 2
	Chapter 2: Derivative and applications	12	0	6	0	30			
3 - 6	- Definition of Derivative - Differentiable Functions - The Chain Rule - Formulas of Derivatives - Second Derivative and Physical Meaning - Higher Derivatives - Logarithmic Differentiation - Absolute and Local Maximum/Minimum Values - Extreme Value Theorem	12	0	6	0	30	[1], [2], [3], [4]		At class - Positive listening, asking questions - Practice as instructed - Doing exercises After class - Do homework Read: - [1] chapters 3 & 4

	- Some example applications of derivative Teaching methods: Lecture, examples, exercises								- [3] chapters 2 & 3
	Chapter 3: Sequences and series	6	0	3	0	15			
7 - 8	- Definition and Properties of Sequences and Series - Monotonic Sequences - Common Series - Convergence and convergence criteria - Absolute Convergence Teaching methods: Lecture, examples, exercises	6	0	3	0	15	[1], [2], [3], [4]		At class - Positive listening, asking questions - Practice as instructed - Doing exercises After class - Do homework Read: - [1] Chapter 10 - [3] Chapter 11
	Chapter 4: Partial derivatives	12	0	6	0	30			
9 - 12	- Functions of More than One Variable and Their Graphs - Partial Derivatives - Directional Derivatives - Tangent Plane - Linear Approximation - Extreme Values - Local Extreme Values - Critical Point - Lagrange Multipliers Teaching methods: Lecture, examples, exercises	12	0	6	0	30	[1], [2], [3], [4]		At class - Positive listening, asking questions - Practice as instructed - Doing exercises After class - Do homework Read: - [1] Chapter 14 - [3] Chapter 14
	Chapter 5: Integral and applications	9	0	3	0	21			
13 - 15	- Definite integral - Geometrical meaning of the integral - Properties of Definite Integrals - Basic theorem of calculus - Indefinite integral	9	0	3	0	21	[1], [2], [3], [4]		At class - Positive listening, asking questions - Practice as instructed

	- Substitution rule - Integration by Parts - Applications of integral Teaching methods: Lecture, examples, exercises								- Doing exercises After class - Do homework Read: - [1] chapters 5 & 6 - [3] chapters 4 & 5
	Total	45	0	30	0	120			

9. Recommended optional programme component:

10. Work placement(s):

11. Date of first approval: October 25, 2022

Course designer

Trịnh Hùng Cường

Reviewer

Nguyễn Quốc Bình

Head of Division

Lê Anh Cường

Dean of faculty

Phạm Văn Huy

12. Date and Up-to-date content

Version: 1.0 - Date issued: 02/26/2020 Version: 2.0 - Date issued: 07/28/2020 Version: 2.1 - Date issued: 07/19/2021 Version: 2.2 - Date issued: 10/22/2022		
Head of Undergraduate Studies Department	Dean of faculty	Instructor